Tutorial - 1

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1 Suppose that :

$A = \{x : x \in N \text{ and } x \text{ is even}\}$

$B = \{x : x \in N \text{ and } x \text{ is prime}\}$

$C = \{x : x \in N \text{ and } x \text{ is a multiple of } 5\}$

Describe following sets

a. $A \cap B$

→ Contains elements that are both even and prime

$A \cap B = \{2\}$

b. $B \cap C$

→ Contains elements that are both prime and multiple of 5 $B \cap C = \{5\}$

c $A \cup B$

→ Contains elements that are either prime or even.

$A \cup B = \{2, 3, 4, 5, 6, 7, 8 \ldots\}$

d $A \cap (B \cup C)$

→ Contains elements that are either prime or multiple of 5

$A \cap (B \cup C) = \{2, 10, 20 \ldots\}$

2. Prove

ii $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

Inclusion 1 : $A \cap (B \cup C) \subseteq (A \cap B) \cup (A \cap C)$

Inclusion 2 : $(A \cap B) \cup (A \cap C) \subseteq A \cap (B \cup C)$

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→ For inclusion 1

Take an arbitrary element $x \in A \cap (B \cup C)$. This means that $x \in A$ and $x \in B \cup C$. From $x \in A \cap (B \cup C)$,

Case 1: If $x \in B$, then $x \in A$, we have, $x \in A \cap B$

so, $x \in (A \cap B) \cup (A \cap C)$

Case 2: If $x \in C$ then $x \in A$, from $x \in A \cap (B \cup C)$ we have $x \in A \cap C$. so, $x \in (A \cup B) \cup (A \cap C)$

Hence, $A \cap (B \cup C) \subseteq (A \cap B) \cup (A \cap C)$.

For inclusion 2

Take an arbitrary element $x \in (A \cap B) \cup (A \cap C)$. This means that $x \in A \cap B$ or $x \in A \cap C$.

Case 1: If $x \in A \cap B$, then $x \in A$ and $x \in B$. Since $x \in B$, it follows that $x \in B \cup C$ because $B \subseteq B \cup C$.

so, $x \in A \cap (B \cup C)$

Case 2: If $x \in A \cap C$, then, $x \in A$ and $x \in C$, since $x \in C$, it follows that $x \in B \cup C$ because $C \subseteq B \cup C$

so, $x \in A \cap (B \cup C)$

Therefore, $\therefore (A \cap B) \cup (A \cap C) \subseteq A \cap (B \cup C)$

$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ proved

3. Define a function
a. That is onto but not one-to-one
b. That is one to one but not onto.

→ One such function is, $f(n) = 2n$ where $f: N \to N$ is defined by multiplying each number by 2.
One to one (Injective):

→ A function is one to one if different inputs map to different outputs. Here, if,
$f(n_1) = f(n_2)$ then, $2n_1 = 2n_2$
which implies, $n_1 = n_2$
Therefore, $f(n) = 2n$ is injective.

Onto (Surjective)

→ A function is onto if every element in the co-domain has a corresponding element in the domain. Here, the co-domain is N (the set of all natural numbers) However, $f(n) = 2n$ only produces even numbers in N, so there is no $n \in N$ that maps to any odd natural number. Therefore, the function $f(n) = 2n$ is not onto.

Conclusion:
The function $f(n) = 2n$ is one to-one but not onto.

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4. The diagonal of matrix A are the entries a_{ij} where $i = j$

a. Write down the 3x3 matrix with ones on diagonal and zeros else where

→ $$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

b. Write down the 3 X 4 matrix with ones on diagonal and zeros elsewhere.

→ $$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

c. Write down the 4x3 matrix with ones on diagonal and zeros elsewhere.

→ $$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

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5.4. Let $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 3 & 4 \end{bmatrix}$ and $D = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 4 \end{bmatrix}$

Compute AD and DA

→ $$AD = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 3 & 4 \end{bmatrix} \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 4 \end{bmatrix}$$

$$= \begin{bmatrix} 2+0+0 & 0+3+0 & 0+0+4 \\ 2+0+0 & 0+6+0 & 0+0+12 \\ 2+0+0 & 0+9+0 & 0+0+16 \end{bmatrix}$$

$$= \begin{bmatrix} 2 & 3 & 4 \\ 2 & 6 & 12 \\ 2 & 9 & 16 \end{bmatrix}$$

$$DA = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 4 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 3 & 4 \end{bmatrix}$$

$$= \begin{bmatrix} 2+0+0 & 2+0+0 & 2+0+0 \\ 0+3+0 & 0+6+0 & 0+9+0 \\ 0+0+4 & 0+0+12 & 0+0+16 \end{bmatrix}$$

$$= \begin{bmatrix} 2 & 2 & 2 \\ 3 & 6 & 9 \\ 4 & 12 & 16 \end{bmatrix}$$

- Hence, AD and DA are transpose of each other

6. Let $A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$, $B = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}$ and $C = \begin{bmatrix} 4 & 3 \\ 0 & 2 \end{bmatrix}$

Verify that $AB = AC$.

$\Rightarrow$ $AB = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}\begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}$

$= \begin{bmatrix} 2+2 & 1+6 \\ 4+4 & 2+12 \end{bmatrix}$

$= \begin{bmatrix} 4 & 7 \\ 8 & 14 \end{bmatrix}$

$AC = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}\begin{bmatrix} 4 & 3 \\ 0 & 2 \end{bmatrix}$

$= \begin{bmatrix} 4+0 & 3+4 \\ 8+0 & 6+8 \end{bmatrix}$

$= \begin{bmatrix} 4 & 7 \\ 8 & 14 \end{bmatrix}$

$AB = AC$ proved

7. Prove that $(AB)^T = B^T A^T$

$\Rightarrow$ let, $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$; $B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$

Here,

$AB = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}\begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$

$$= \begin{bmatrix} 5+14 & 6+16 \\ 15+28 & 18+32 \end{bmatrix}$$

$$= \begin{bmatrix} 19 & 22 \\ 43 & 50 \end{bmatrix} \quad (AB)^T = \begin{bmatrix} 19 & 43 \\ 22 & 50 \end{bmatrix}$$

$$B^TA^T = \begin{bmatrix} 5 & 7 \\ 6 & 8 \end{bmatrix} \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}$$

$$= \begin{bmatrix} 5+14 & 15+28 \\ 6+16 & 18+32 \end{bmatrix}$$

$$= \begin{bmatrix} 19 & 43 \\ 22 & 50 \end{bmatrix}$$

$$(AB)^T = B^TA^T$$

8 Let A be a rectangular matrix given by $A = \begin{bmatrix} a & d \\ b & e \\ c & f \end{bmatrix}$. Compute A^TA and show that it is a symmetric square matrix. Observe that the sum of the diagonal elements of A^TA is the sum of square of the elements of A.

→ $A = \begin{bmatrix} a & d \\ b & e \\ c & f \end{bmatrix} \quad A^T = \begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix}$

$$A^TA = \begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix} \begin{bmatrix} a & d \\ b & e \\ c & f \end{bmatrix}$$

$$= \begin{bmatrix} a^2+b^2+c^2 & ad+be+cf \\ ad+be+cf & d^2+e^2+f^2 \end{bmatrix}$$

diagonal of A^T : $a^2+b^2+c^2+d^2+e^2+f^2$

Sum of square of element of $A = \begin{bmatrix} a & d \\ b & e \\ c & f \end{bmatrix}$

is $a^2+b^2+c^2+d^2+e^2+f^2$

9. Find the Inverse of matrices $A = \begin{bmatrix} 5 & 6 \\ 4 & 5 \end{bmatrix}$

and $B = \begin{bmatrix} 6 & 4 \\ 3 & 3 \end{bmatrix}$

$\rightarrow$ $D_A = \begin{vmatrix} 5 & 6 \\ 4 & 5 \end{vmatrix} = 25 - 24 = 1$

$$A^{-1} = \frac{1}{D_A} \begin{bmatrix} 5 & -6 \\ -4 & 5 \end{bmatrix}$$

$$= \frac{1}{1} \begin{bmatrix} 5 & -6 \\ -4 & 5 \end{bmatrix}$$

$$= \begin{bmatrix} 5 & -6 \\ -4 & 5 \end{bmatrix}$$

$D_B = \begin{vmatrix} 6 & 4 \\ 3 & 3 \end{vmatrix} = 18 - 12 = 6$

$$B^{-1} = \frac{1}{D_B} \begin{bmatrix} 3 & -4 \\ -3 & 6 \end{bmatrix}$$

$$= \frac{1}{6}\begin{bmatrix} 3 & -4 \\ -3 & 6 \end{bmatrix}$$

$$= \begin{bmatrix} 1/2 & -2/3 \\ -1/2 & 1 \end{bmatrix}$$

10 Given matrices A and B below, verify that they are inverses.

$$A = \begin{bmatrix} 4 & 1 \\ 3 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & -1 \\ -3 & 4 \end{bmatrix}$$

→ $$AB = \begin{bmatrix} 4 & 1 \\ 3 & 1 \end{bmatrix}\begin{bmatrix} 1 & -1 \\ -3 & 4 \end{bmatrix}$$

$$= \begin{bmatrix} 4-3 & -4+4 \\ 3-3 & -3+4 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$BA = \begin{bmatrix} 1 & -1 \\ -3 & 4 \end{bmatrix}\begin{bmatrix} 4 & 1 \\ 3 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 4-3 & 1-1 \\ -12+12 & -3+4 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$AB = BA = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Both are inverses of each other.

11. Verify that given matrices are inverses of each other.

$$A = \begin{bmatrix} 1 & -1 & 0 \\ 1 & 0 & -1 \\ 2 & 3 & -4 \end{bmatrix} \quad B = \begin{bmatrix} 3 & -4 & 1 \\ 2 & -4 & 1 \\ 3 & -5 & 1 \end{bmatrix}$$

$\rightarrow$

$$AB = \begin{bmatrix} 1 & -1 & 0 \\ 1 & 0 & -1 \\ 2 & 3 & -4 \end{bmatrix} \begin{bmatrix} 3 & -4 & 1 \\ 2 & -4 & 1 \\ 3 & -5 & 1 \end{bmatrix}$$

$$= \cancel{\begin{bmatrix} 3+4+0 & -4+4+0 \end{bmatrix}}$$

$$= \begin{bmatrix} 3-2+0 & -4+4+0 & 1-1+0 \\ 3+0-3 & -4+0+5 & 1+0-1 \\ 6+6-12 & -8-12+20 & 2+3-4 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$BA = \begin{bmatrix} 3 & -4 & 1 \\ 2 & -4 & 1 \\ 3 & -5 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 & 0 \\ 1 & 0 & -1 \\ 2 & 3 & -4 \end{bmatrix}$$

$$= \begin{bmatrix} 3-4+2 & -3+0+3 & 0+4-4 \\ 2-4+2 & -2+0+3 & 0+4-4 \\ 3-5+2 & -3+0-3 & 0+5-4 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$AB = BA$, so they are inverse of each other.

12. For apple, bananas and cherries distributed into 3 baskets

1 In basket 1; 2 apples, 3 bananas, 1 cherry with weight of 12g

2. In basket 2; 1 apple, 2 bananas, 4 cherries with weight of 10g

3 In basket 3; 3 apples, 1 banana, 2 cherries with weight of 14g

let, a is weight of one apple
b " " " " banana
c " " " " cherry

Tasks:

- Set up a system of equations based on the information provided
- Represent the system of equations in matrix form.

→ Equations:

$$2a + 3b + c = 12$$
$$a + 2b + 4c = 10$$
$$3a + b + c = 14$$

Matrix form:

$$\begin{bmatrix} 2 & 3 & 1 \\ 1 & 2 & 4 \\ 3 & 1 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 12 \\ 10 \\ 14 \end{bmatrix}$$

13 Adult tickets = \$15 sold tickets = 250
student tickets = \$10 total money = \$2825
child tickets = \$8
No. of student tickets sold is twice the adult ones.
let;

x = no. of ticket sold to adult at \$15
y = " " " " " student " \$10
z = " " " " " child " \$8

a. set up equations
b. Represent in matrix form

→ Equations

$$x + y + z = 250$$
$$15x + 10y + 8z = 2825$$

we have, $y = 2x$

$$x + 2x + z = 250 \quad / \quad 3x + z = 250 \text{ — (i)}$$
$$15x + 10(2x) + 8z = 2825 \quad / \quad 35x + 8z = 2825 \text{ — (ii)}$$
$$\text{and } y = 2x \text{ — (iii)}$$

Matrix form,

$$\begin{bmatrix} 3 & 0 & 1 \\ 35 & 0 & 8 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 250 \\ 2825 \end{bmatrix}$$

Equations;

$$x + y + z = 250$$
$$15x + 10y + 8z = 2825$$
$$y = 2x \;/\; -2x + y + 0z = 0$$

Matrix form;

$$\begin{bmatrix} 1 & 1 & 1 \\ 15 & 10 & 8 \\ -2 & 1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 250 \\ 2825 \\ 0 \end{bmatrix}$$

14

→ t = cost of one shirt
v = cost of video
s = cost of stuffed animal

$$4t + v + s = 135$$
$$2t + v + 3s = 115$$
$$2t + v + s = 85$$

Matrix form,

$$\begin{bmatrix} 4 & 1 & 1 \\ 2 & 1 & 3 \\ 2 & 1 & 1 \end{bmatrix} \begin{bmatrix} t \\ v \\ s \end{bmatrix} = \begin{bmatrix} 135 \\ 115 \\ 85 \end{bmatrix}$$

15

→ x = measure of first angle
y = " " second "
z = " " third "

→ $x + y + z = 180$

$y + z = 2x \;/\; -2x + y + z = 0$

$z = y + 12 \;/\; 0x - y + z = 12$

Matrix form,

$$\begin{bmatrix} 1 & 1 & 1 \\ -2 & 1 & 1 \\ 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 180 \\ 0 \\ 12 \end{bmatrix}$$

16

→ c = cost of one pound of candy

k = " " " box of cookies

p = " " " can of popcorn

$2c + 3k + p = 65$

$4c + 6k + 3p = 140$

$8c + 8k + 5p = 250$

Matrix form,

$$\begin{bmatrix} 2 & 3 & 1 \\ 4 & 6 & 3 \\ 8 & 8 & 5 \end{bmatrix} \begin{bmatrix} c \\ k \\ p \end{bmatrix} = \begin{bmatrix} 65 \\ 140 \\ 250 \end{bmatrix}$$

17 Solving linear equations using Matrix method

a. $\begin{cases} x + 2y = 4 \\ 3x - 5y = 1 \end{cases}$

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- let, $A = \begin{bmatrix} 1 & 2 \\ 3 & -5 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 4 \\ 1 \end{bmatrix}$

$AX = B \quad / \quad X = BA^{-1}$

$|A| = \begin{vmatrix} 1 & 2 \\ 3 & -5 \end{vmatrix} = |-5-6| = -11$

$A^{-1} = \frac{1}{-11} \begin{bmatrix} -5 & -2 \\ -3 & 1 \end{bmatrix}$

Now,

$X = BA^{-1}$

$\begin{bmatrix} x \\ y \end{bmatrix} = \frac{1}{11} \begin{bmatrix} 4 \\ 1 \end{bmatrix} \begin{bmatrix} -5 & -2 \\ -3 & 1 \end{bmatrix}$

$= \frac{-1}{11} \begin{bmatrix} -20-2 \\ -12+1 \end{bmatrix}$

$= \begin{bmatrix} -22/-11 \\ -11/11 \end{bmatrix}$

$= \begin{bmatrix} 2 \\ 1 \end{bmatrix}$

Hence, $x = 2$, $y = 1$

b $\begin{cases} 5x + y = 13 \\ 3x + 2y = 5 \end{cases}$

→ let, $A = \begin{bmatrix} 5 & 1 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 13 \\ 5 \end{bmatrix}$

$AX = B \quad / \quad X = BA^{-1}$

$$DA = \begin{vmatrix} 5 & 1 \\ 3 & 2 \end{vmatrix} = (10-3) = 7$$

$$A^{-1} = \frac{1}{7}\begin{bmatrix} 2 & -1 \\ -3 & 5 \end{bmatrix}$$

Now,

$$X = BA^{-1} \quad 2\times 1 \quad 2\times 2$$

$$= \frac{1}{7}\begin{bmatrix} 13 \\ 5 \end{bmatrix}\begin{bmatrix} 2 & -1 \\ -3 & 5 \end{bmatrix}$$

$$\cancel{= \frac{1}{7}\begin{bmatrix} 26 & 13\times 2 + 13\times -3 & 13\times -1 + 13\times 5 \\ 5\times 2 + & & \end{bmatrix}}$$

$$= \frac{1}{7}\begin{bmatrix} 13\times 2 + 5\times -1 \\ 13\times -3 + 5\times 5 \end{bmatrix}$$

$$= \frac{1}{7}\begin{bmatrix} 26 - 5 \\ -39 + 25 \end{bmatrix}$$

$$= \frac{1}{7}\begin{bmatrix} 21 \\ -14 \end{bmatrix}$$

$$= \begin{bmatrix} 21/7 \\ -14/7 \end{bmatrix}$$

$$= \begin{bmatrix} 3 \\ -2 \end{bmatrix}$$ Hence, $x = 3$, $y = -2$

c $\begin{cases} 3x + 2y = -2 \\ x + 4y = 6 \end{cases}$

$\rightarrow$ let $A = \begin{bmatrix} 3 & 2 \\ 1 & 4 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -2 \\ 6 \end{bmatrix}$

$$D_A = \begin{vmatrix} 3 & 2 \\ 1 & 4 \end{vmatrix} = |12-2| = 10$$

$$A^{-1} = \frac{1}{10}\begin{bmatrix} 4 & -2 \\ -1 & 3 \end{bmatrix}$$

Now,

$$AX = B$$

$$X = BA^{-1}$$

$$= \frac{1}{10}\begin{bmatrix} -2 \\ 6 \end{bmatrix}\begin{bmatrix} 4 & -2 \\ -2 & 3 \end{bmatrix}$$

$$= \frac{1}{10}\begin{bmatrix} -8-12 \\ 2+18 \end{bmatrix}$$

$$= \frac{1}{10}\begin{bmatrix} -20 \\ 20 \end{bmatrix}$$

$$= \begin{bmatrix} -2 \\ 2 \end{bmatrix}$$

Hence, $x = -2$ and $y = 2$

d. $\begin{cases} 2x + 3y = 6 \\ x - y = 1/2 \end{cases}$

→ let, $A = \begin{bmatrix} 2 & 3 \\ 1 & -1 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 6 \\ 1/2 \end{bmatrix}$

$$AX = B \;/\; X = BA^{-1}$$

$$D_A = \begin{vmatrix} 2 & 3 \\ 1 & -1 \end{vmatrix} = |-2-3| = -5$$

$$A^{-1} = \frac{1}{-5}\begin{bmatrix} -1 & -3 \\ -1 & 2 \end{bmatrix}$$

Now

$$X = BA^{-1}$$

$$= \frac{1}{-5}\begin{bmatrix} 6 \\ 1/2 \end{bmatrix}\begin{bmatrix} -1 & -3 \\ -1 & 2 \end{bmatrix}$$

$$= \frac{1}{-5}\begin{bmatrix} -6 & -3/2 \\ -6 & +2/2 \end{bmatrix}$$

$$= \frac{1}{-5}\begin{bmatrix} -15/2 \\ \cancel{-28}/5 \end{bmatrix}$$

$$= \begin{bmatrix} 3/2 \\ 1 \end{bmatrix}$$

$x = 3/2$ and $y = 1$

e $\begin{cases} 4x + 3y = 11 \\ x - 3y = -1 \end{cases}$

$\rightarrow$ let $A = \begin{bmatrix} 4 & 3 \\ 1 & -3 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 11 \\ -1 \end{bmatrix}$

$AX = B$ / $X = BA^{-1}$

$$DA = \begin{vmatrix} 4 & 3 \\ 1 & -3 \end{vmatrix} = |-12 - 3| = -15$$

$$A^{-1} = \frac{1}{-15}\begin{bmatrix} -3 & -3 \\ -1 & 4 \end{bmatrix}$$

Now $X = BA^{-1}$

$$= \frac{-1}{15}\begin{bmatrix} 11 \\ -1 \end{bmatrix}\begin{bmatrix} -3 & -3 \\ -1 & 4 \end{bmatrix}$$

$$= \frac{-1}{15}\begin{bmatrix} -33 + 3 \\ -11 - 4 \end{bmatrix}$$

$$= \frac{-1}{15}\begin{bmatrix} -30 \\ -15 \end{bmatrix}$$

$$= \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

Hence, $x = 2,\ y = 1$

f. $\begin{cases} 2x + 3y = 8 \\ 4x + 6y = 20 \end{cases}$

$\rightarrow$ let, $A = \begin{bmatrix} 2 & 3 \\ 4 & 6 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 8 \\ 20 \end{bmatrix}$

$AX = B$

$D_A = \begin{vmatrix} 2 & 3 \\ 4 & 6 \end{vmatrix} = |12 - 12| = 0$

Not possible.

g. $\begin{cases} x + y - z = 2 \\ x + z = 7 \\ 2x + y + z = 13 \end{cases}$

$\rightarrow$ let $A = \begin{bmatrix} 1 & 1 & -1 \\ 1 & 0 & 1 \\ 2 & 1 & 1 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 \\ 7 \\ 13 \end{bmatrix}$

$AX = B \,/\, X = BA^{-1}$

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$$|A| = \begin{vmatrix} 1 & 1 & -1 \\ 1 & 0 & 1 \\ 2 & 1 & 1 \end{vmatrix}$$

$$= 1\begin{vmatrix} 0 & 1 \\ 1 & 1 \end{vmatrix} - 1\begin{vmatrix} 1 & 1 \\ 2 & 1 \end{vmatrix} - 1\begin{vmatrix} 1 & 0 \\ 2 & 1 \end{vmatrix}$$

$$= -1 + 1 - 1$$

$$= -1$$

Cofactor of $a_{11} = \begin{vmatrix} 0 & 1 \\ 1 & 1 \end{vmatrix} = |0-1| = -1$

cofactor of $a_{12} = -\begin{vmatrix} 1 & 1 \\ 2 & 1 \end{vmatrix} = -|1-2| = 1$

cofactor of $a_{13} = \begin{vmatrix} 1 & 0 \\ 2 & 1 \end{vmatrix} = |1-0| = 1$

cofactor of $a_{21} = -\begin{vmatrix} 1 & -1 \\ 1 & 1 \end{vmatrix} = -|1+1| = -2$

cofactor of $a_{22} = \begin{vmatrix} 1 & -1 \\ 2 & 1 \end{vmatrix} = |1+2| = 3$

cofactor of $a_{23} = -\begin{vmatrix} 1 & 1 \\ 2 & 1 \end{vmatrix} = -|1-2| = 1$

cofactor of $a_{31} = \begin{vmatrix} 1 & -1 \\ 0 & 1 \end{vmatrix} = |1-0| = 1$

cofactor of $a_{32} = -\begin{vmatrix} 1 & -1 \\ 1 & 1 \end{vmatrix} = -|1+1| = -2$

cofactor of $a_{33} = \begin{vmatrix} 1 & 1 \\ 1 & 0 \end{vmatrix} = |0-1| = -1$

cofactor of $A = \begin{bmatrix} -1 & 1 & 1 \\ -2 & 3 & 1 \\ 1 & -2 & -1 \end{bmatrix}$

Adjucate $= \begin{bmatrix} -1 & -2 & 1 \\ 1 & 3 & -2 \\ 1 & 1 & -1 \end{bmatrix}$

Now,

$$X = BA^{-1}$$

$$= \begin{bmatrix} 2 \\ 7 \\ 13 \end{bmatrix} \begin{bmatrix} 1 & 2 & -1 \\ -1 & -3 & 2 \\ -1 & -1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 2 + 14 - 13 \\ -2 - 21 + 26 \\ -2 - 7 + 13 \end{bmatrix}$$

$$= \begin{bmatrix} 3 \\ 3 \\ 4 \end{bmatrix}$$

$x = 3,\ y = 3,\ z = 4$

h.g

$$\begin{cases} x + y - z = 2 \\ 3x + y = 7 \\ x + y + 2z = 3 \end{cases}$$

→ let $A = \begin{bmatrix} 1 & 1 & -1 \\ 3 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 \\ 7 \\ 3 \end{bmatrix}$

$$AX = B$$

$$D_A = \begin{vmatrix} 1 & 1 & -1 \\ 3 & 1 & 0 \\ 1 & 1 & 2 \end{vmatrix}$$

$$= 1\begin{vmatrix} 1 & 0 \\ 1 & 2 \end{vmatrix} - 1\begin{vmatrix} 3 & 0 \\ 1 & 2 \end{vmatrix} - 1\begin{vmatrix} 3 & 1 \\ 1 & 1 \end{vmatrix}$$

$$= 2 - 6 - 2$$

$$= -6$$

Cofactor of $a_{11} = \begin{vmatrix} 1 & 0 \\ 1 & 2 \end{vmatrix} = 2$

cofactor of $a_{12} = -\begin{vmatrix} 3 & 0 \\ 1 & 2 \end{vmatrix} = -6$

cofactor of $a_{13} = \begin{vmatrix} 3 & 1 \\ 1 & 1 \end{vmatrix} = 2$

cofactor of $a_{21} = -\begin{vmatrix} 1 & -1 \\ 1 & 2 \end{vmatrix} = -3$

cofactor of $a_{22} = \begin{vmatrix} 1 & -1 \\ 1 & 2 \end{vmatrix} = 3$

cofactor of $a_{23} = -\begin{vmatrix} 1 & 1 \\ 1 & 1 \end{vmatrix} = 0$

cofactor of $a_{31} = \begin{vmatrix} 1 & -1 \\ 1 & 0 \end{vmatrix} = 1$

cofactor of $a_{32} = -\begin{vmatrix} 1 & -1 \\ 3 & 0 \end{vmatrix} = -3$

cofactor of $a_{33} = \begin{vmatrix} 1 & 1 \\ 3 & 1 \end{vmatrix} = -2$

cofactor of A $= \begin{bmatrix} 2 & -6 & 2 \\ -3 & 3 & 0 \\ 1 & -3 & -2 \end{bmatrix}$

Adjugate of A $= \begin{bmatrix} 2 & -3 & 1 \\ -6 & 3 & -3 \\ 2 & 0 & -2 \end{bmatrix}$

$X = BA^{-1}$

$= -\frac{1}{6} \begin{bmatrix} 2 \\ 7 \\ 3 \end{bmatrix} \begin{bmatrix} 2 & -3 & 1 \\ -6 & 3 & -3 \\ 2 & 0 & -2 \end{bmatrix}$

$= -\frac{1}{6} \begin{bmatrix} 4 - 21 + 3 \\ -12 + 21 - 9 \\ 4 + 0 - 6 \end{bmatrix}$

$= -\frac{1}{6} \begin{bmatrix} 14 \\ 0 \\ -2 \end{bmatrix}$

$= \begin{bmatrix} -7/3 \\ 0 \\ 1/3 \end{bmatrix}$

i. $\begin{cases} -x - 2y + z = 1 \\ 2x + 3y = 2 \\ y - 2z = 0 \end{cases}$

$\rightarrow$ $\begin{bmatrix} -1 & -2 & 1 \\ 2 & 3 & 0 \\ 0 & 1 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}$

$= \left[\begin{array}{ccc|c} -1 & -2 & 1 & 1 \\ 2 & 3 & 0 & 2 \\ 0 & 1 & -2 & 0 \end{array}\right]$

$= R_2 \rightarrow R_2 + 2R_1$

$\left[\begin{array}{ccc|c} -1 & -2 & 1 & 1 \\ 0 & -1 & 2 & 4 \\ 0 & 1 & -2 & 0 \end{array}\right]$

$R_3 \rightarrow R_3 + R_2$

$= \left[\begin{array}{ccc|c} -1 & -2 & 1 & 1 \\ 0 & -1 & 2 & 4 \\ 0 & 0 & 0 & 4 \end{array}\right]$

Inconsistent equation.

j. $\begin{cases} x + 4y - z = 4 \\ 2x + 5y + 8z = 15 \\ x + 3y - 3z = 1 \end{cases}$

$\rightarrow$ $\begin{bmatrix} 1 & 4 & -1 \\ 1 & 5 & 8 \\ 1 & 3 & -3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 4 \\ 15 \\ 1 \end{bmatrix}$

$$= \left[\begin{array}{ccc|c} 1 & 4 & -1 & 4 \\ 2 & 5 & 8 & 15 \\ 1 & 3 & -3 & 1 \end{array}\right]$$

$R_2 \to R_2 - 2R_1$

$$= \left[\begin{array}{ccc|c} 1 & 4 & -1 & 4 \\ 0 & -3 & 10 & 7 \\ 1 & 3 & -3 & 1 \end{array}\right]$$

$R_3 \to R_3 - R_1$

$$= \left[\begin{array}{ccc|c} 1 & 4 & -1 & 4 \\ 0 & -3 & 10 & 7 \\ 0 & -1 & -2 & -3 \end{array}\right]$$

$R_3 \to 3R_3 - R_2$

$$= \left[\begin{array}{ccc|c} 1 & 4 & -1 & 4 \\ 0 & -3 & 10 & 7 \\ 0 & 0 & -16 & -16 \end{array}\right]$$

$-16z = -16$ / $-3y + 10 = 7$ / $x + 4 - 1 = 4$

$\therefore z = 1$ / $\therefore y = 1$ / $\therefore x = 1$

K.

$$\begin{cases} 5x + 3y + 9z = -1 \\ -2x + 3y - z = -2 \\ -x - 4y + 5z = 1 \end{cases}$$

$$\rightarrow \left[\begin{array}{ccc|c} 5 & 3 & 9 & -1 \\ -2 & 3 & -1 & -2 \\ -1 & -4 & 5 & 1 \end{array}\right]$$

$R_2 \rightarrow R_2 - 2R_3$

$$= \left[\begin{array}{ccc|c} 5 & 3 & 9 & -1 \\ 0 & 11 & -11 & -4 \\ -1 & -4 & 5 & 1 \end{array}\right]$$

$R_2 \rightarrow R_2 + \frac{2}{5} R_1$

$$\left[\begin{array}{ccc|c} 5 & 3 & 9 & -1 \\ 0 & 21/5 & -23/5 & -13/5 \\ -1 & -4 & 5 & 1 \end{array}\right]$$

$R_3 \rightarrow 5R_3 + R_1$

$$= \left[\begin{array}{ccc|c} 5 & 3 & 9 & -1 \\ 0 & 1 & -11 & -4 \\ 0 & -17 & 34 & 4 \end{array}\right]$$

$R_3 \rightarrow R_3 + \frac{1}{5} R_1$

$$= \left[\begin{array}{ccc|c} 5 & 3 & 9 & -1 \\ 0 & 21/5 & -23/5 & -13/5 \\ 0 & -17/5 & 46/5 & 4/5 \end{array}\right]$$

$R_2 \rightarrow R_2 / 21/5$

$$= \left[\begin{array}{ccc|c} 5 & 3 & 9 & -1 \\ 0 & 1 & -23/21 & -13/21 \\ 0 & -17/5 & 46/5 & 4/5 \end{array}\right]$$

$R_3 \rightarrow R_3 + 17/5 R_2$, $R_1 \rightarrow R_1 - 3R_2$

$$= \left[\begin{array}{ccc|c} 5 & 0 & 216/21 & 26/21 \\ 0 & 1 & -23/21 & -13/21 \\ 0 & 0 & 529/105 & 44/21 \end{array}\right]$$

$R_3 \rightarrow R_3 / 529/105$

$$= \left[\begin{array}{ccc|c} 5 & 0 & 216/21 & 26/21 \\ 0 & 1 & -23/21 & -13/21 \\ 0 & 0 & 1 & 0 \end{array}\right]$$

$R_1 \rightarrow R_1 - 216/21 R_3$

$= R_2 \rightarrow R_2 + 23/21 R_3$

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 0 \end{array}\right]$$

6 $\begin{cases} 6x - 5y + 2z = 3 \\ 2x + y - 4z = 5 \\ 3x - 3y + z = -1 \end{cases}$

$$= \begin{bmatrix} 6 & -5 & 2 & 3 \\ 2 & 1 & -4 & 5 \\ 3 & -3 & 1 & -1 \end{bmatrix}$$

$= R_1 \to R_1/6$

$$\left(\begin{array}{ccc|c} 1 & -5/6 & 1/3 & 1/2 \\ 2 & 1 & -4 & 5 \\ 3 & -3 & 1 & -1 \end{array}\right)$$

$= R_3 \to R_3 - 3R_1$

$$\left(\begin{array}{ccc|c} 1 & -5/6 & 1/3 & 1/2 \\ 0 & 16/6 & -14/3 & 9/2 \\ 0 & -13/6 & 0 & -5/2 \end{array}\right)$$

$= R_2 \to R_2 / 16/6 = R_2 - 6/16$

$$\left(\begin{array}{ccc|c} 1 & -5/6 & 1/3 & 1/2 \\ 0 & 1 & -14/8 & 27/16 \\ 0 & -13/6 & 0 & -5/2 \end{array}\right)$$

$= R_3 \to R_3 + 13/6 R_2$

$$\begin{pmatrix} 1 & -5/6 & 1/3 & 1/2 \\ 0 & 1 & -14/8 & 27/16 \\ 0 & 0 & 1 & 2 \end{pmatrix}$$

$= R_1 \to R_1 + 5/6 R_2$

$$\left(\begin{array}{ccc|c} 1 & 0 & 1/2 & 7/4 \\ 0 & 1 & -7/4 & 27/16 \\ 0 & 0 & 1 & 2 \end{array}\right)$$

$R_1 \to R_1 - 1/2 R_3$

$$\left(\begin{array}{ccc|c} 1 & 0 & 0 & 4 \\ 0 & 1 & -7/4 & 27/16 \\ 0 & 0 & 1 & 2 \end{array}\right)$$

$R_2 \to R_2 + 7/4 R_3$

$$\left(\begin{array}{ccc|c} 1 & 0 & 0 & 4 \\ 0 & 1 & 0 & 5 \\ 0 & 0 & 1 & 2 \end{array}\right)$$

18. Confirm why the following system has no solution:

$$\begin{cases} x + 2y - 3z = -1 \\ x - 3y + z = 1 \\ 2x - y - 3z = 2 \end{cases}$$

$$\left[\begin{array}{ccc|c} 1 & 2 & -3 & -1 \\ 1 & -3 & 1 & 1 \\ 1 & -1 & -3 & 2 \end{array}\right]$$

$R_2 \to R_2 - R_1$

$$\left[\begin{array}{ccc|c} 1 & 2 & -3 & -1 \\ 0 & -5 & 4 & 2 \\ 2 & 1 & -3 & 2 \end{array}\right]$$

$R_3 \to R_3 - 2R_1$

$$\left[\begin{array}{ccc|c} 1 & 2 & -3 & -1 \\ 0 & -5 & 4 & 2 \\ 0 & -5 & 3 & 4 \end{array}\right]$$

$R_3 \to R_3 - R_2$

$$\left[\begin{array}{ccc|c} 1 & 2 & -3 & -1 \\ 0 & -5 & 4 & 2 \\ 0 & 0 & -1 & 2 \end{array}\right]$$

$R_3 \to -R_3$

$$\left[\begin{array}{ccc|c} 1 & 2 & -3 & -1 \\ 0 & -5 & 4 & 2 \\ 0 & 0 & 1 & -2 \end{array}\right]$$

$R_2 \to R_2 - 4R_3$

$$\left[\begin{array}{ccc|c} 1 & 2 & -3 & -1 \\ 0 & -5 & 0 & 10 \\ 0 & 0 & 1 & -2 \end{array}\right]$$

$R_2 \to R_2 / -5$

$$\left[\begin{array}{ccc|c} 1 & 2 & -3 & -1 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & -2 \end{array}\right]$$

$R_1 \to R_1 - 2R_2 + 3R_3$

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & -3 \\ 0 & 1 & 0 & -3 \\ 0 & 0 & 1 & -2 \end{array}\right]$$

$x = -3$

$y = -2$

$z = -2$ unique solution

19. $\begin{cases} 2x - 2y + 3z = 6 \\ 4x - 3y + 2z = 0 \\ -2x + 3y - 7z = 1 \end{cases}$

$\rightarrow \left[\begin{array}{ccc|c} 2 & -2 & 3 & 6 \\ 4 & -3 & 2 & 0 \\ -2 & 3 & -7 & 1 \end{array}\right]$

$\because R_1 \rightarrow R_1/2$

$\left[\begin{array}{ccc|c} 1 & -1 & 3/2 & 3 \\ 4 & -3 & 2 & 0 \\ -2 & 3 & -7 & 1 \end{array}\right]$

$R_2 \rightarrow R_2 - 4R_1$, $R_3 \rightarrow R_3 + 2R_1$

$\left[\begin{array}{ccc|c} 1 & -1 & 3/2 & 3 \\ 0 & 1 & -4 & -12 \\ 0 & 1 & -4 & -5 \end{array}\right]$

$\cdot\ R_3 \rightarrow R_3 - R_2$

$\left[\begin{array}{ccc|c} 1 & -1 & 3/2 & 3 \\ 0 & 1 & -4 & 12 \\ 0 & 0 & 0 & 7 \end{array}\right]$

Third row implies $0 = 7$, which is a contradiction. Therefore, the system has no solution because the eqn represent planes that do not intersect

20. $\begin{cases} x + 2y - z = 1 \\ 2x + 7y + 4z = 11 \\ x + 3y + z = 4 \end{cases}$

$\rightarrow \left[\begin{array}{ccc|c} 1 & 2 & -1 & 1 \\ 2 & 7 & 4 & 11 \\ 1 & 3 & 1 & 4 \end{array}\right]$

$R_2 \rightarrow R_2 - 2R_1 \quad R_3 \rightarrow R_3 - R_1$

$\left[\begin{array}{ccc|c} 1 & 2 & -1 & 1 \\ 0 & 3 & 6 & 9 \\ 0 & 1 & 2 & 3 \end{array}\right]$

$R_2 \rightarrow R_2/3$

$\left[\begin{array}{ccc|c} 1 & 2 & -1 & 1 \\ 0 & 1 & 2 & 3 \\ 0 & 1 & 2 & 3 \end{array}\right]$

$R_3 \rightarrow R_3 - R_2$

$\left[\begin{array}{ccc|c} 1 & 2 & -1 & 1 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 0 \end{array}\right]$

The third row is $0 = 0$, indicating dependence. The remaining two eqn can be parameterized. Assign $z = t$, then solve for x and y in terms of t.

21. find $f'(x)$

a. $f(x) = x + \sqrt{x}$

→ $f'(x) = \dfrac{d(x + \sqrt{x})}{dx}$

$= \dfrac{dx}{dx} + \dfrac{d\sqrt{x}^{1/2}}{dx}$

$= 1 + \dfrac{1}{2} x^{1/2 - 1}$

$= 1 + 1/2\, x^{-1/2}$

$= 1 + \dfrac{1}{2\sqrt{x}}$

b. $f(x) = 2 + 1/x + 1/x^2$

→ $f'(x) = \dfrac{d\,(2 + 1/x + 1/x^2)}{dx}$

$= \dfrac{d\,2}{dx} + \dfrac{dx^{-1}}{dx} + \dfrac{dx^{-2}}{dx}$

$= 0 - x^{-2} - 2x^{-3}$

$= \dfrac{-1}{x^2} - \dfrac{2}{x^3}$

c. $f(x) = \sqrt{x}\cos x$

→ $f'(x) = u\,dv + v\,du$

$= \cancel{-\sqrt{x}\sin x}$

$= \sqrt{x}\,\dfrac{d\cos x}{dx} + \cos x\,\dfrac{d\sqrt{x}}{dx}$

$= -\sqrt{x}\sin x + \cos x \, \frac{1}{2} x^{1/2 - 1}$

$= \frac{1}{2\sqrt{x}} \cos x - \sqrt{x}\sin x$

d. $f(x) = x^3 e^x$

$\rightarrow$ $f'(x) = u\,dv + v\,du$

$= x^3 \frac{de^x}{dx} + e^x \frac{dx^3}{dx}$

$= e^x x^3 + 3e^x x^2$

$= e^x (x^3 + 3x^2)$

e. $f(x) = \frac{x^3}{e^x}$

$\rightarrow$ $f'(x) = \frac{u\,dv + v\,du}{v^2}$

$= \frac{x^3 de^x/dx + e^x dx^3/dx}{(e^x)^2}$

$= \frac{\cancel{e^x}(x^3 + 3x^2)}{\cancel{e^x} \cdot e^x}$

$= \frac{3x^2 + x^3}{e^x}$

f $f(x) = (3x^2 + 2x + 1)^5$

$\rightarrow$ $f'(x) = \frac{d(3x^2 + 2x + 1)^5}{dx}$

$$= \frac{d(3x^2+2x+1)^5}{d(3x^2+2x+1)} \times \frac{d(3x^2+2x+1)}{dx}$$

$$= (3x^2+2x+1)(6x+2)$$

g. $f(x) = (\sin x)^4$

$\Rightarrow f'(x) = \frac{d(\sin x)^4}{dx}$

$$= \frac{d(\sin x)^4}{d\sin x} \times \frac{d\sin x}{dx}$$

$$= 4\sin^3 x \times \cos x$$

$$= 4\sin^3 x \cos x$$

h. $f(x) = e^{(3x^2+2x)}$

$\Rightarrow f'(x) = \frac{d e^{(3x^2+2x)}}{dx}$

$$= \frac{d e^{(3x^2+2x)}}{d(3x^2+2x)} \times \frac{d(3x^2+2x)}{dx}$$

$$= e^{(3x^2+2x)} \times (6x+2)$$

$$= (6x+2)e^{(3x^2+2x)}$$

i. $f(x) = \ln(5x^2+3x)$

$\Rightarrow f'(x) = \frac{d\ln(5x^2+3x)}{dx}$

$$= \frac{d\ln(5x^2+3x)}{d(5x^2+3x)} \times \frac{d(5x^2+3x)}{dx}$$

$$= \frac{1}{5x^2+3x} \times (10x+3)$$

$$= \frac{10x+3}{5x^2+3x}$$

j. $f(x) = \sqrt{x^3+4x+1}$

$\rightarrow$ $f'(x) = \frac{d(\sqrt{x^3+4x+1})}{dx}$

$$= \frac{d(x^3+4x+1)^{1/2}}{d(x^3+4x+1)} \times \frac{d(x^3+4x+1)}{dx}$$

$$= \frac{1}{2}(x^3+4x+1)^{-1/2} \quad (3x^2+4)$$

$$= \frac{3x^2+4}{2\sqrt{x^3+4x+1}}$$

22. Suppose that $g(x) = x^2 - 3x + 2$ and $f(x)$ is a differentiable function. All we know about $f(x)$ is the following:

- $f(0) = 3$ and $f'(0) = -1$
- $f(1) = 5$ and $f'(1) = 0$
- $f(2) = -2$ and $f'(2) = 3$
- $f(3) = 6$ and $f'(3) = 1$

If possible:

1. Find the derivatives of $f(x) \cdot g(x)$ at $x = 1$

$\rightarrow$ $g(x) = x^2 - 3x + 2$

$g(1) = (1)^2 - 3 \times 1 + 2 = 0$

$g'(x) = 2x - 3$

$g'(1) = 2 \times 1 - 3 = -1$

We have,

$f(1) = 5$

$f'(1) = 0$

$g(1) = 0$

$g'(1) = -1$

Now,

$$\frac{d(f(x) \cdot g(x))}{dx} = f(x)g'(x) + g(x)f'x$$

At $x = 1$, $= f(1)g'(1) + g(1)f(1)$

$= 5 \times -1 + 0 \times 0$

$= -5$

2 Find the derivatives of $\frac{f(x)}{g(x)}$ at $x = 0$

→ $g(x) = x^2 - 3x + 2$

$g(0) = (0)^2 - 3 \times 0 + 2 = 2$

$g'(x) = 2x - 3$

$g'(0) = 2 \times 0 - 3 = -3$

We have,

$f(0) = 3$

$f'(0) = -1$

$g(0) = 2$

$g'(0) = -3$

Now,

$$\frac{d\left(f(x)/g(x)\right)}{dx} = \frac{f(x)\,g'(x) + g(x)\,f'(x)}{(g(x))^2}$$

At, $x = 0$, $= \dfrac{-f(0)\,g'(0) + g(0)\,f'(0)}{(g(0))^2}$

$$= \cancel{\frac{3\times 3 + (-3)\times(-1)}{(-3)^2}} \; \frac{(3\times -3) + 2\times -1}{(2)^2}$$

$$= \cancel{\frac{9-1}{4}} = \frac{9-2}{4}$$

$$= \cancel{3/4 \; 9/9} = 7/4$$

3. Find the derivative of $f(g(x))$ at $x = 0$

⇒ At $x = 0$,

$f'(0) = -1$

$f(0) = 3$

$g'(0) = -3$

$g(0) = 2$

$f'(2) = 3$

Now,

$$f(g(x)) = f'(g(x))\,g'(x)$$

At, $x = 0$, $= f'(g(0))\,g'(0)$

$$= f'(2) \times (-3)$$

$$= 3 \times -3$$

$$= -9$$

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23 For the following functions find the maximum or minimum of the function at given interval using critical points

a. $g(x) = x^2 - 4x + 5$ at interval $[1, 4]$

→ Here,

$$g'(x) = 2x - 4$$

For critical points;

$$g'(x) = 0$$

or, $2x - 4 = 0$

or, $2(x - 2) = 0$

$\therefore x = 2$

$g(2) = (2)^2 - 4\times2 + 5 = 4 - 8 + 5 = 1$

$g(1) = (1)^2 - 4\times1 + 5 = 1 - 4 + 5 = 2$

$g(4) = (4)^2 - 4\times4 + 5 = 16 - 16 + 5 = 5$

Therefore;

Minimum at $g(2) = 1$ and Maximum at $g(4) = 5$

b. $h(x) = \frac{x^2}{2} - 2x$ at interval $[-2, 3]$

→ Here,

$$h'(x) = \frac{2\times1}{2}x - 2 = x - 2$$

For critical points,

$$h'(x) = 0$$

$x - 2 = 0$ / $x = 2$

$$h(2) = \frac{(2)^2}{2} - 2\times 2$$

$$= 1 - 4$$

$$= -2$$

$$h(-2) = \frac{(-2)^2}{2} - 2\times -2$$

$$= 2 + 4$$

$$= 6$$

$$h(3) = \frac{(3)^2}{4} - 2\times 3$$

$$= \frac{9}{4} - 6$$

$$= \frac{9-24}{4}$$

$$= -15/4$$

Therefore;

Maximum at $h(-2) = 6$

Minimum at $h(2) = -2$

c $p(x) = 3x^3 - 9x^2 + 7$ at interval $[-1, 2)$

→ Here;

$p'(x) = 9x^2 - 18x$

For critical points,

$$9x^2 - 18x = 0$$

$$9x(x - 2) = 0$$

$x = 2$

$p(x) = 3x^3 - 9x^2 + 7$

$p(2) = 3 \times (2)^3 - 9 \times (2)^2 + 7$

$= 3 \times 8 - 9 \times 4 + 7$

$= 24 - 36 + 7$

$= 31 - 36$

$= -5$

$p(-1) = 3 \times (-1)^3 - 9 \times (-1)^2 + 7$

$= -3 - 9 + 7$

$= -12 + 7$

$= -5$

$p(2) = 3 \times (2)^3 - 9 \times (2)^2 + 7$

$= 24 - 36 + 7$

$= 31 - 36$

$= -5$

$p(0) = 3 \times (0)^3 - 9 \times (0)^2 + 7$

$= 7$

Therefore;

Minimum at $p(-1 \text{ or } 2) = -5$

Maximum at $p(0) = 7$

d. $f(x) = x^2 - 4x + 6$ at interval $[0, 3]$

→ Here,

$f'(x) = 2x - 4$

For critical points;

$2x - 4 = 0$

$2(x - 2) = 0$ / $x = 2$

$f(2) = (2)^2 - 4\times2 + 6$
$= 4 - 8 + 6$
$= 2$

$f(0) = (0)^2 - 4\times0 + 6$
$= 6$

$f(3) = (3)^2 - 4\times3 + 6$
$= 9 - 12 + 6$
$= 15 - 12$
$= 3$

Therefore,
Minimum at $f(2) = 2$
Maximum at $f(0) = 6$

c. $g(x) = -2x^3 + 3x^2 + 12x$ at interval $[-1, 2]$

→ Here,
$g'(x) = -6x^2 + 6x + 12$
for critical point,
$-6x^2 + 6x + 12 = 0$
or $6(-x^2 + x + 2) = 0$
or, $x^2 - x - 2 = 0$
or $x^2 - 2x + x - 2 = 0$
or $x(x-2) + 1(x-2) = 0$
or $(x-2)(x+1) = 0$
$\therefore x = 2, x = -1$

$g(-1) = -2\times(-1)^3 + 3\times(-1)^2 + 12\times-1$

$= 2 + 3 - 12$

$= -7$

$g(2) = -2 \times (2)^3 + 3 \times (2)^2 + 12 \times 2$

$= -16 + 12 + 24$

$= 20$

Therefore;

Minimum at $g(-1) = -7$

Maximum at $g(2) = 20$

24. Evaluate the gradient of a function at given points.

1. $f(x,y) = x^2 + y^2$ at:

a. $(0,0)$

$\rightarrow \frac{df}{dx} = \frac{d(x^2+y^2)}{dx}$

$= 2x + y^2$

$\frac{df}{dy} = \frac{d(x^2+y^2)}{dy}$

$= x^2 + 2y$

$\nabla f = \begin{pmatrix} df/dx \\ df/dy \end{pmatrix}$

$= \begin{pmatrix} 2x + y^2 \\ x^2 + 2y \end{pmatrix}$

At, $(0,0)$,

$$\nabla f(0,0) = \begin{pmatrix} 2\times 0 + (0)^2 \\ (0)^2 + 2\times 0 \end{pmatrix}$$

$$= \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

b) $(1,3)$

$$\rightarrow \nabla f(1,3) = \begin{pmatrix} 2\times 1 + (3)^2 \\ (1)^2 + 2\times 3 \end{pmatrix}$$

$$= \begin{pmatrix} 11 \\ 7 \end{pmatrix}$$

b. $(1,3)$

$$\rightarrow \nabla f(1,3) = \begin{pmatrix} 2\times 1 \\ 2\times 3 \end{pmatrix}$$

$$= \begin{pmatrix} 2 \\ 6 \end{pmatrix}$$

c. $(-1,-5)$

$$\rightarrow \nabla f(-1,-5) = \begin{pmatrix} 2\times -1 \\ 2\times -5 \end{pmatrix}$$

$$= \begin{pmatrix} -2 \\ -10 \end{pmatrix}$$

2. $f(x, y, z) = x^3 z - 2y^2 x + 5z$ at,

a. $(1, 1, -4)$

$\rightarrow$ $\frac{df}{dx} = \frac{d(x^3 z - 2y^2 x + 5z)}{dx}$

$= 3x^2 z - 2y^2$

$\frac{df}{dy} = \frac{d(x^3 z - 2y^2 x + 5z)}{dy}$

$= -4yx$

$\frac{df}{dz} = \frac{d(x^3 z - 2y^2 x + 5z)}{dz}$

$= x^3 + 5$

Now,

$$\nabla f(1, 1, -4) = \begin{pmatrix} 3x^2 z - 2y^2 \\ -4yx \\ x^3 + 5 \end{pmatrix}$$

$$= \begin{pmatrix} 3 \times (1)^2 \times -4 - 2 \times (1)^2 \\ -4 \times 1 \times 1 \\ (1)^3 + 5 \end{pmatrix}$$

$$= \begin{pmatrix} -14 \\ -4 \\ 6 \end{pmatrix}$$

b. $(0, 1, 0)$

$\rightarrow$ $\nabla f(0, 1, 0) = \begin{pmatrix} 3 \times (0)^2 \times (0)^2 - 2 \times (1)^2 \\ -4 \times 1 \times 0 \\ (0)^3 + 5 \end{pmatrix}$

$$= \begin{pmatrix} -2 \\ -9 \\ 5 \end{pmatrix}$$

c. $(-3, -2, 1)$

$\rightarrow \nabla f(-3,-2,1) = \begin{pmatrix} 3\times(-3)^2\times 1 - 2\times(-2)^2 \\ -4\times -2\times -3 \\ (-3)^3+5 \end{pmatrix}$

$$= \begin{pmatrix} 19 \\ -24 \\ -22 \end{pmatrix}$$

3. $m(x,y) = \ln(x^2+y^2+1)$ at:

a. $(0,0)$

$\rightarrow \frac{dm}{dx} = \frac{d\ln(x^2+y^2+1)}{d(x^2+y^2+1)} \times \frac{d(x^2+y^2+1)}{dx}$

$= \frac{1}{x^2+y^2+1} \times 2x$

$= \frac{2x}{x^2+y^2+1}$

$\frac{dm}{dy} = \frac{d\ln(x^2+y^2+1)}{d(x^2+y^2+1)} \times \frac{d(x^2+y^2+1)}{dy}$

$= \frac{2y}{x^2+y^2+1}$

Now,

$\nabla m(x,y) = \begin{pmatrix} \frac{2x}{x^2+y^2+1} \\ \frac{2y}{x^2+y^2+1} \end{pmatrix}$

$$= \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

b. $(1,-1)$

$\rightarrow$ $\nabla m(1,-1) = \begin{pmatrix} \frac{2x}{x^2+y^2+1} \\ \frac{2y}{x^2+y^2+1} \end{pmatrix} = \begin{pmatrix} \frac{2\times 1}{(1)^2+(-1)^2+1} \\ \frac{2\times -1}{(1)^2+(-1)^2+1} \end{pmatrix}$

$$= \begin{pmatrix} 2/3 \\ -2/3 \end{pmatrix}$$

c. $(-2,3)$

$\rightarrow$ $\nabla m(-2,3) = \begin{pmatrix} \frac{2\times -2}{(-2)^2+(3)^2+1} \\ \frac{2\times 3}{(-2)^2+(3)^2+1} \end{pmatrix} = \begin{pmatrix} -4/14 \\ 6/14 \end{pmatrix}$

$$= \begin{pmatrix} -2/7 \\ 3/7 \end{pmatrix}$$

25. For a given function, evaluate the maximum and minimum values over the specified interval:

1. $m(x,y) = x^2+y^2-4x-6y+13$ interval $x = [0,3]$ and $y = [0,4]$.

$\rightarrow$ $\frac{df}{dx} = 2x-4$

$$\frac{df}{dy} = 2y - 6$$

Gradient;

$$\nabla f(x,y) = \begin{pmatrix} 2x-4 \\ 2y-6 \end{pmatrix}$$

For critical points,

Gradient $= 0$

So, $2x - 4 = 0 \;/\; x = 2$

$2y - 6 = 0 \;/\; y = 3$

Critical points $= (2,3)$

Now,

Check if the critical point $(2,3)$ is with in interval $[1,3] \times [1,4]$

$$f(2,3) = 2^2 + 3^2 - 4\times 2 - 6\times 3 + 13$$
$$= 0$$

Here,

For $(1,1) = (1)^2 + (1)^2 - 4\times 1 - 6\times 1 + 13 = 5$

For $(1,4) = (1)^2 + (4)^2 - 4\times 1 - 6\times 4 + 13 = 2$

For $(3,1) = (3)^2 + (1)^2 - 4\times 3 - 6\times 1 + 13 = 5$

For $(4,4) = (3)^2 + (4)^2 - 4\times 3 - 6\times 4 + 13 = 2$

And, we have, $(2,3) = 0$

Therefore;

Maximum is 5 at $(1,1)$ and minimum is 0 at $(2,3)$.

2. $g(x,y) = xy + x^2 - 4y + 6$ interval $x = [-2,2]$ and $y = [1,3]$

$\rightarrow \frac{dg}{dx} = y + 2x$

$\frac{dg}{dy} = x - 4$

Gradient,

$\nabla g(x,y) = \begin{pmatrix} 2x+y \\ x-4 \end{pmatrix}$

For critical points,

Gradient $= 0$

so, $2x + 4 + y = 0$

$y = -8$

$x - 4 = 0$

$x = 4$

$g(4,-8) = 22$

$g(-2,1) = 4$

$g(-2,3) = -8$

$g(2,1) = 8$

$g(2,3) = 4$

3) $p(x,y) = x^3 - 3xy + 4$ interval $x[0,2]$ and $[0,3]$

$\rightarrow \nabla p(x,y) = \begin{pmatrix} 3x^2 - 3y \\ -3x \end{pmatrix}$

$3x^2 - 3y = 0$ | $-3x = 0$

$x^2 - y = 0$ | $x = 0$

$p(0,0) = 4$

$p(0,3) = 4$

$p(2,0) = 8 - 0 + 4 = 12$

$p(2,3) = -6$

minimum at $(2,3) = -6$

maximum at $(2,0) = 12$

4. $h(x,y) = 2x^2 + 3y^2 - 12x - 18y + 20$

$x = [1,5]$ $y = [0,4]$

$\rightarrow \nabla h(x,y) = \begin{pmatrix} 4x - 12 \\ 6y - 18 \end{pmatrix}$

$4x - 12 = 0$ | $6y - 18 = 0$

$x = 3$ | $y = 3$

$h(3,3) = -25$

$h(1,0) = 10$

$h(1,4) = -14$

$h(8,0) = 10$

$h(8,4) = -14$

maximum $= 10$ $(1,0)$ $(8,0)$

minimum $= -25$ $(3,3)$